and so on. AMD's future APUs are speculated to have a combination of x86 and ARM cores. Discrete GPUs may also use HSA to access system memory.

The variety of players in the mobile devices segment such as ARM, Qualcomm and Samsung, among others, shows how relevant this technology is for mobile devices. HSA-compliant applications will greatly improve efficiency which will translate into more tasks getting done in lesser time. As far as mobile devices are concerned, this equates to improved battery performance.

HSA solution stack

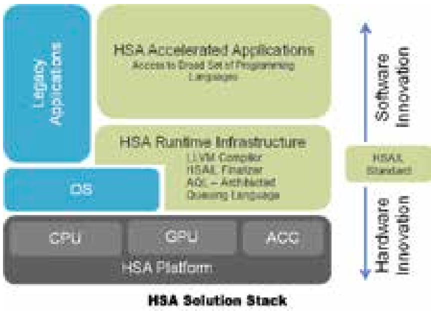
The HSA Solution Stack allows multiple hardware solutions to be exposed to software using a common low-level interface layer called the HSA Intermediate Language (HSAIL). It acts as a one-stop shop for low-level software and tools. HSAIL frees the programmer from having to program for specific platforms i.e. The same code can run on different CPU/GPU configurations. HSA will allow developers to program at a higher abstraction level using mainstream programming languages such as C++, Python and Java and using libraries that target HSA.

The HSA Foundation announced its first specification by releasing a Version 0.95 of its Programmer's Reference Manual. More details are available here: http://goo.gl/80Lqn2

Amongst the founding members, so far, only AMD has showcased a working prototype of the technology in action. Let's take a look at AMD's proposed implementation of HSA which it calls the Heterogeneous Unified Memory Access (hUMA). But first, let's check out the HSA Solution Stack.

AMD hUMA

AMD has impressed us with the performance of its graphics cores on its accelerated processing units (APUs) since the Llano APU to the current Richland APU. The next generation of APUs code-named 'Kaveri', which will enter the market towards the end of this year or in early 2014, will sport the hUMA. In the current set-up, as far as memory is concerned, the integrated GPU partitions off a section of the memory for itself. The CPU has no access to this memory. With hUMA technology, the CPU and GPU will share a single memory space so that there will be no pointless back and forth copying of data. Pointers can be used to access data that has been processed either by the CPU or GPU, and the GPU can take advantage of the CPU virtual address space. Thanks to this ease of accessing data between the CPU and GPU, application developers will no longer have to code functions twice.



HSAIL frees the programmer from having to program for specific platforms

The three main features of hUMA are as follows:

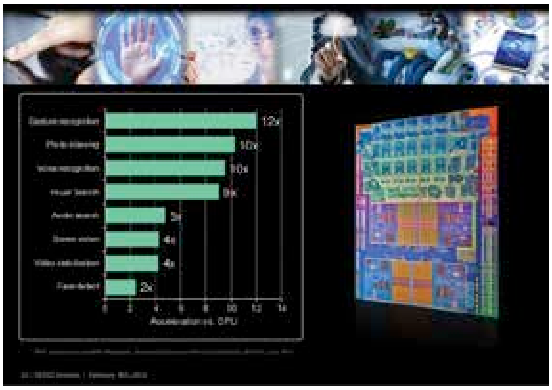
- **Bi-directional coherent memory:** This implies that the changes or updates made by any one processing element will be seen by all the other processing elements simultaneously.
- **Pageable memory:** Currently, GPUs cannot handle page faults like the CPU can, but with HSA GPUs will be able to and it also removes restricted page-locked memory.
- **Access to entire memory space:** As can be seen in the image alongside, the CPU and GPU memory pools are separate but with HSA, both the CPU and GPU will be given access to the entire memory space. This includes virtual and physical memory address spaces.

With HSA, the GPU and the CPU will be given access to the same memory

Applications

AMD envisions the impact of HSA to be felt in various areas such as augmented reality, biometric recognition, natural UI and gestures, multi-point HD video conferencing and beyond HD content streaming and transcoding. Sure, getting the application developers to make programs HSA compliant is a big challenge which AMD will have to face. The chart below will give you a rough idea of how much of an improvement HSA can bring to certain present-day tasks. Of course till we see this in action, the figures below should be taken with a pinch of salt.

AMD is working with LibreOffice, an Open Source office suite, to allow it to make use of GPU computing for accelerating spreadsheet calculations. This will come as a relief to users of the Calc application (similar to MS Excel) who feel that calculations with large data sets is slow. One of the features of the LibreOffice 4.1 suite is its HSA compliancy. LibreOffice aims to change the order of calculations to make them more efficient and will convert tasks to OpenCL so that they can be run on GPUs.



Application acceleration using HSA

As far as the console market is concerned, AMD has its sights on both, the PlayStation 4 and Xbox One, the next-gen consoles coming out later this year. Both of them sport the AMD APU and the architecture is HSA compliant. The PlayStation 4 has 8GB of GDDR5 memory which is addressable by the CPU and the GPU.